

CDC 2026

AUTHOR
Ari Staggs

PUBLISHED
September 26, 2026

Packages

```
library(tidyverse)
library(knitr)
library(tidymodels)
library(rms) #calculate VIF
library(car)
```

Load Data

```
music <- read_csv("C:/Users/Ari/Downloads/music.csv")
```

Create Training and Test Data

```
set.seed(101)
sample_index <- sample(1:nrow(music), size = floor(0.85*nrow(music)))

train_music <- music[sample_index,]
test_music <- music[-sample_index,]
```

Create Models

```
anxiety_model <- lm(anxiety~classical+country+edm+folk+gospel+hiphop+jazz+kpop+latin+lofi+metal+pop+rnb+rock)
depression_model <- lm(depression~classical+country+edm+folk+gospel+hiphop+jazz+kpop+latin+lofi+metal+pop+rnb+rock)
insomnia_model <- lm(insomnia~classical+country+edm+folk+gospel+hiphop+jazz+kpop+latin+lofi+metal+pop+rnb+rock)
ocd_model <- lm(ocd~classical+country+edm+folk+gospel+hiphop+jazz+kpop+latin+lofi+metal+pop+rnb+rock)

tidy(anxiety_model) |> filter(p.value<0.05)
```

A tibble: 1 × 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
1 (Intercept)	4.93	0.321	15.4	3.35e-45

```
tidy(depression_model) |> filter(p.value<0.05)
```

```
# A tibble: 3 × 5
  term          estimate std.error statistic  p.value
  <chr>         <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)    3.25      0.345      9.41 9.85e-20
2 country       -0.717     0.297     -2.41 1.62e- 2
3 rock           0.924     0.316      2.93 3.56e- 3
```

```
tidy(insomnia_model) |> filter(p.value<0.05)
```

```
# A tibble: 2 × 5
  term          estimate std.error statistic  p.value
  <chr>         <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)    2.84      0.357      7.95 8.86e-15
2 metal          1.02      0.282      3.63 3.11e- 4
```

```
tidy(ocd_model) |> filter(p.value<0.05)
```

```
# A tibble: 1 × 5
  term          estimate std.error statistic  p.value
  <chr>         <dbl>     <dbl>     <dbl>   <dbl>
1 (Intercept)    2.49      0.333      7.48 2.68e-13
```

Check for Multicollinearity

```
vif(anxiety_model)
```

```
classical  country      edm      folk    gospel    hiphop    jazz    kpop
1.115808  1.114720  1.107487  1.199986  1.098588  2.026876  1.196087  1.161413
  latin    lofi    metal      pop      rnb      rap      rock      vgm
1.117682  1.228652  1.263389  1.192610  1.508286  1.986248  1.365860  1.150393
```

Predict Values for Test Data

```
anxiety_pred <- predict(anxiety_model, newdata = test_music)
depression_pred <- predict(depression_model, newdata = test_music)
insomnia_pred <- predict(insomnia_model, newdata = test_music)
ocd_pred <- predict(ocd_model, newdata = test_music)

test_music <- test_music |> mutate(anxiety_pred=anxiety_pred,depression_pred=depression_pred,insomnia_pred=insomnia_pred)
```

Assess Fit

```
sqrt(mean((test_music$anxiety_pred-test_music$anxiety)^2))
```

```
[1] 2.922004
```

```
sd(test_music$anxiety)
```

```
[1] 2.893659
```

```
sqrt(mean((test_music$depression_pred-test_music$depression)^2))
```

```
[1] 3.017475
```

```
sd(test_music$depression)
```

```
[1] 3.108362
```

```
sqrt(mean((test_music$insomnia_pred-test_music$insomnia)^2))
```

```
[1] 3.080118
```

```
sd(test_music$insomnia)
```

```
[1] 3.12793
```

```
sqrt(mean((test_music$oed_pred-test_music$oed)^2))
```

```
[1] 2.937339
```

```
sd(test_music$oed)
```

```
[1] 2.883196
```